

The dependence of energy and momentum on fiducial 4-vector fields

P.G.L. Porta Mana 

Western Norway University of Applied Sciences <pgl@portamana.org>

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1 Questions about energy

The present note discusses two sets of questions that at first sight might not appear to be connected, also because one set is typical of Newtonian mechanics and appears already at an undergraduate level, while the other appears very naturally in general relativity. But it will be shown that the questions of the two sets are instead deeply connected and their answers can be traced to a common idea.

- A1. What is the total energy of an extended body? Should it include internal, kinetic, gravitational potential energy? or only some of these?
- A2. Why is internal energy the same in all frames, whereas kinetic energy is frame-dependent?
- A3. Why is it possible to derive physical laws about energy by consider “virtual displacements”?
- A4. Is Bernoulli’s theorem a consequence of the balance of energy? or of the balance of momentum?
- B1. How can we associate a total energy to a region of spacetime?